# NexaVoice (Salesforce AI Calling Agent)

## Project Completion & Status Report

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## Executive Summary

The NexaVoice project is currently estimated to be **85% - 90% complete**. The core structural foundations across both the frontend mobile application (Flutter) and the AI/Backend (FastAPI) are fully implemented. The application has been successfully rebranded, the AI processing pipelines (Deepgram + Gemini) are functional, and the DevOps configurations (Docker) are ready for deployment. The project is now entering its final integration phase, requiring live credentials and end-to-end network testing over the public internet.

## Section 1: Work Completed (100%)

### *1.1 Frontend App & Branding (Flutter)*

- Executed complete NexaVoice rebranding using the Emerald Green (#1B8B4B) & Charcoal color palette.
- Integrated the official NexaVoice app icon and logo assets.
- Redesigned the Login Screen with smooth animations and premium UI cards.
- Upgraded Dashboard, Transcripts, and Call Job cards with gradient pills and priority visualization.
- Built the Supervisor Console for seamlessly syncing leads from Salesforce to the local database.

### *1.2 AI Engine & Backend (FastAPI)*

- Integrated Deepgram Live API for real-time speech-to-text transcription.
- Integrated Google Gemini AI to handle intelligent call scripting and dynamic conversations.
- Created the Call Orchestrator logic to manage bidirectional audio streaming between the caller and the AI.
- Developed telephony API endpoints mapping out Exotel, Airtel SIP, and Twilio integration architectures.

### *1.3 Database & DevOps Architecture*

- Defined complete PostgreSQL database schema via Alembic for Users, Leads, Calls, and Devices.
- Created Dockerfile and docker-compose.yml for production-ready AWS deployment.
- Completed detailed Telephony Provider Cost Analysis predicting rates as low as ■1.39/min (Airtel SIP).

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# Section 2: Pending Work (The Final 10% - 15%)

## *2.1 Salesforce Production Credentials*

**Current Status:** The backend uses mock fallback data for testing.
**Action Required:** Input live Salesforce API credentials (Client ID, Secret, Username, Password) into the backend environment variables so real leads can be synced.

## *2.2 Live Telephony Testing (Exotel / Airtel)*

**Current Status:** The endpoints for SIP routing are structurally built but await live network access.
**Action Required:** Acquire a live Exotel Account (or Airtel SIP Trunk), configure the API keys, and perform end-to-end live testing over the public internet to ensure audio streams successfully flow into Deepgram and Gemini.

## *2.3 App Offline/Network Resilience*

**Current Status:** The Flutter app assumes the backend is always reachable. If the backend goes down, the app hangs with an infinite loading spinner.
**Action Required:** Update the frontend networking logic (DioClient & Providers) to catch connection errors and gracefully display a 'Backend Offline' UI overlay instead of crashing.

## *2.4 Gradle & Build Upgrades*

**Current Status:** The Android APK builds successfully but triggers deprecation warnings from older Gradle/Kotlin plugins.
**Action Required:** Upgrade Gradle to version 8.14.0 and resolve plugin warnings to ensure long-term App Store compliance and maintainability.

## *2.5 AWS / Cloud Deployment*

**Current Status:** Docker files are prepared, but the app is currently only running on localhost/Wi-Fi.
**Action Required:** Spin up an AWS EC2 instance, deploy the Docker containers, configure Nginx + SSL, and update the Flutter AppConfig to point to the new public domain name.